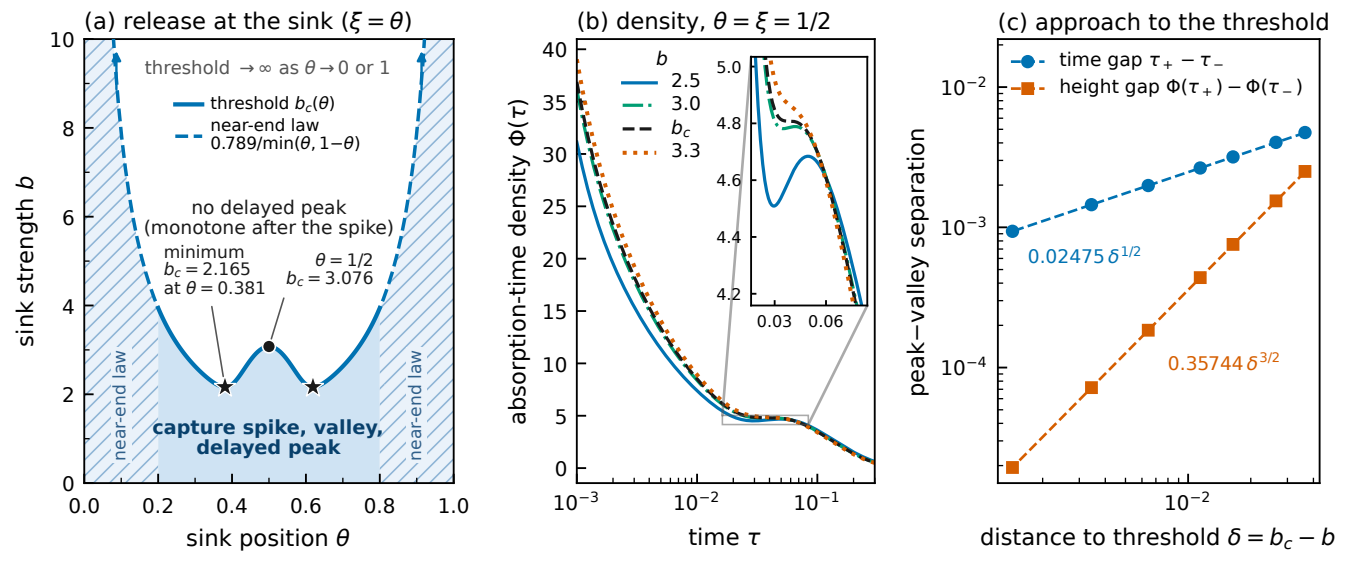
(a) release at the sink ($\xi = \theta$)
threshold $\to \infty$ as $\theta \to 0$ or 1
threshold $b_c(\theta)$
near-end law
$0.789/\min(\theta, 1-\theta)$
no delayed peak
(monotone after the spike)
minimum
$b_c = 2.165$
at $\theta = 0.381$
$\theta = 1/2$
$b_c = 3.076$
near-end law
near-end law
capture spike, valley,
delayed peak
sink strength b
sink position θ
(b) density, $\theta = \xi = 1/2$
b
2.5
3.0
b_c
3.3
absorption-time density $\Phi(\tau)$
time τ
(c) approach to the threshold
time gap $\tau_+ - \tau_-$
height gap $\Phi(\tau_+) - \Phi(\tau_-)$
$0.02475\,\delta^{1/2}$
$0.35744\,\delta^{3/2}$
peak−valley separation
distance to threshold $\delta = b_c - b$